

The Algorithm of Speed Dating

a. Abstract

Speed Dating began as a turn of the century social experiment more than anything. Participant's sign up to meet possible romantic partners under a time constraint with surveys to quantitatively analyze their interactions with each other- in hopes of a paper love calculation turned real life connection. In this article, the authors explore the relationships, factors, and components that can be analyzed to find linear explanations in the speed-dating data by using Principal Component Analysis, Canonical Correlation Analysis, Factor Analysis, Correspondence Analysis, Lasso Regression, and Ridge Regression. Exploring the data through multiples of these multivariate statistical techniques reinforced and expanded upon the complex and multidimensional relationships that can be derived from the experiments survey answers. In many ways, the findings in this study match popular preconceptions and echoed disparaging sentiments that physical attraction is a driving force in a successful date. The data does repeatedly show that beautiful people do, in fact, more often have more success in speed-dating as in many endeavors. However, the data also clearly indicates that the relationships between one's perception of themselves can forecast their future partnership potential. The models revealed that, while physical attraction is important, it is made up of many unique factors, and those factors are at least partially in the participant's own perception of themselves- underscoring that confidence is key to success in successful speed-dating experiences.

b. Introduction

In recent years, speed-dating was revived as an alternative to online dating apps. Single people of all ages have experienced two decades of scamming, catfishing, stalking, cheating, and risk-inherent "modern" dating techniques. Especially in the post-pandemic, much-needed, overcorrection of physically socializing with other humans, speed-dating will appeal to anyone who feels especially lonely and out of range of meeting someone to build a romantic connection with due limited social interactions during covid restrictions. The design is also meant to curve the risk to the favor of a participant, after all these are other like-minded single folks who crave the same connection and belonging. Speed-dating attracts the busy professional, but also the enlightened loner. It has been commended as a way to "put yourself out there" and meet and mingle with people normally outside of one's typical social circles. The statistical nature of the surveys taken during speed dating events offers daters' imperceptible or underlying insights that help them to find a data-driven connection.

c. Literature Review

Speed Dating Data has provided significant insights into the magic of attraction for many prior researchers, likely due to the nature of the fullness and randomness of speed-dating surveys that is absolutely necessary for the experiment to have a chance at success. The participants are willing and enthusiastic to provide data which is very beneficial for collection of the data. Finkel et al. highlights these participants' immediate judgments of their date's personality, appearance, and chemistry showing their relationships with attraction (Finkel et al.,

2007). This aligns with Asendorpf et al. who explores predicting success in short-term and long-term dating outcomes based on attraction factors. Their research examines attractiveness and sociosexuality emphasizing that in speed-dating, willingness to engage in sex outside of a committed relationship was a significant predictor of the outcome of the date (Asendorpf et al., 2011). Luo et al. reinforce these findings by demonstrating that physical attraction is a dominant factor, but reciprocity, and having things in common play crucial roles in longer term relationships, which is important as a key predictor for a speed-dater's success (Luo et al., 2009). Expanding on these ideas Pierson et al. expands the analysis of factors to include the conversational data and social standing of the participant's as predictors. Their study goes further than the initial physical and chemical attraction, and analyzes the relationship that social behaviors and conversational skills can significantly predict outcomes of speed-dates. Examining this aspect for speed-daters offers a fuller explanation of how to succeed beyond being physically attractive (Pierson et al., 2015). In practical ways, these studies collectively provide a foundation of understanding for speed-daters, that focusing on their self-esteem, conversational skills, and improving their social standings, can set them apart beyond physical attraction and lead them to successful dating outcomes.

d. Methods

This paper draws on six key questions that together capture the dynamics of successful matches in speed dating, with an appropriate statistical method as a means for conducting this analysis. Specifically the research focuses on:

Which underlying factors are most influential in determining the outcomes of speed dating events? Analyzed with Principal Component Analysis (PCA.)

PCA can be used to uncover what is most critical in the outcome of speed dating. The first principal component represented 7.87% of the total variation and emphasized such traits as fun, ambitious, and attractive. While it took 43 components to describe 80% of the total variance, which says something about the complexity of human interactions. Most relevant personality traits that were vital for success concerned being fun and ambitious. Attractiveness did count a lot, too. From the scree plot, it is clearly shown that the first three components had been most important; after the fourth and fifth, their impact decreased, showing the personality of one individual tends to be more important compared with other factors. The findings indicate that personality matters in establishing a good connection during speed dating. **Figure 1**

What's the relationship between self-rating attractiveness and successful speed-dating outcomes? Analyzed with Canonical Correlation Analysis (CCA)

The canonical correlation analysis begins with deciding which variables are correlated with each other in such a way that would make sense to group them into the same canon, or variate, for multivariate linear regression amongst each other. To prepare the data for exploratory analysis for two variates, incomplete data points were removed, and resulted in 7459 complete observations for analysis. The CCA model was trained on 70% of the data, and 30% was used for testing. The canonical correlation variates were found to have met the assumptions for linear regressions and by computing the loadings of the variates and correlations, identified the relationship between the self-rated traits and the dater's outcomes. The Wilks' Lambda was used to test for significance of the variates, and visualizations to show the meaningful relationships in a domain context.

What factors are most important when it comes to speed dating? Analyzed with Factor Analysis (FA)

A scree plot was used to determine that either four or five factors would be appropriate for factor analysis. After running the factor analysis multiple times and removing the variables that did not meet the cutoff value, I was left with four factors and a cutoff value of 0.4. This combination of the number of factors and cutoff value resulted in every variable being loaded into exactly one factor. I received a KMO score of 0.83, which means there is an adequate sample for this dataset and factor analysis was appropriate to use for this data. I used Bartlett's Test of Sphericity and received a value of 69,079 with a significant p-value, which means there is variance in the data. I used Cronbach's Alpha to determine the consistency of the data and received a value of 0.17. This is not a good value and shows that the data may be inconsistent, and this could potentially be an issue with the analysis, but the issue is likely with the structure of the data and is likely unavoidable.

How do participant demographics, preferences, and perceived attributes relate to the likelihood of forming a match during speed dating events? Analyzed with Correspondence Analysis (CA)

The relationships between the participants' characteristics, preferences, and perceived attributes were analyzed using CA, focusing on how those characteristics influence a match's outcome in a speed-dating context. The analysis visually depicted that attractiveness and age are the most influential predictors of a successful match, while other demographic characteristics of the respondents, such as race and occupation, are far weaker. The test of significance of the Chi-square justified that the mentioned associations are significant, hence giving more validity to the results.

What variables strongly influence overall satisfaction with the people met during a speed dating event? Analyzed with Lasso Regression

For this study we decided to use a combination of lasso regression, forward regression and linear regression to understand what variables strongly influence overall satisfaction with the people met during speed dating. Each of these regressions are suitable to work together because the explanatory variable, `satis_2`, is numeric. For this study we started with 195 total variables and 194 x variables. By choosing to use lasso regression to start we were able to minimize the amount of x variables down to 110 x variables. We also found by checking the log lambda output of how many variables we should end with from lasso regression in lasso figure 1. Then from this output we then continued the path of minimizing the model to create a usable model that explains what variables influence overall satisfaction with people met during a speed dating event. We did this by taking the output of the lasso regression and running forward regression to continue to minimize the variables that influence our explanatory variable `satis_2`. From this output we then checked for multicollinearity by checking the VIF values, we found many variables that had multicollinearity. One by one we removed them till we had a non multicollinear model. Once we had this, we then performed linear regression and only kept variables that passed the F test by having p values smaller than 0.05. This output also highlights that the adjusted r square of the final model was 0.443 meaning the model could explain 44.3 percent variance of the y using the x variables.

What factors influence an individual's self-perception of attractiveness in the dating context? Analyzed with Ridge Regression

Ridge was specifically used for its ability to handle multicollinearity in the data set, an issue when predictors are highly correlated. Due to variables in self-perception, preferences in partners, and demographics being interrelated, ridge regression may be preferred over an ordinary least squares model. Ridge regression uses a penalty parameter, lambda, to shrink coefficients and reduce variance to improve model generalizability. All predictors were standardized with a mean=0 and a sd=1. The analysis was conducted over a sequence of lambda values ranging from 0 to 10 with a step size of .1. An illustration of the plot shows how the MSE changes with changing lambda values.

After fitting a final regression model, several diagnostic plots were generated to check for the validity in the model. A Q-Q plot of residuals and histogram were generated to assess whether the residuals followed a normal distribution. The points on the plot primarily follow the diagonal line with slight tails which suggests that the assumption of normality was reasonably met and the histogram depicts a normal distribution curve. A residuals vs. fitted values plot was examined to check for nonlinearity or heteroscedasticity. Residuals are scattered around the horizontal axis without any major patterns, further suggesting no violations of nonlinearity.

e. Discussion and Results

Principal Component Analysis (PCA)

In summary, PCA gives us an opportunity to narrow down research to the most important components which in our case are: **how fun the person is and how ambitious they are**; with attractiveness being in fourth place PCA allows us to understand that speed dating is not just about looks. Personality traits are key factors. This challenges the common belief that appearance is everything. Instead, who you are and how you present yourself to make the biggest effect on your date.

Besides understanding these traits, the insights gained from PCA suggest practical applications. For example, by designing a pre-dating survey that keeps these major personality traits in mind while arranging the participants, organizers will have far more effective speed dates. There is a very good chance that this will improve the success rates of the matches and minimize the number of people leaving the session with the belief that they have wasted their evening. Imagine matching participants based on traits found to be most influential in a match; this might turn speed dating into an even faster, wiser, and more efficient process.

Canonical Correlation Analysis (CCA)

In Canonical Correlation Analysis, three canonical correlations were found and the First Canonical Correlation: 0.79 has a very strong relationship. Because the first canonical variate explains a strong positive relationship between self-rated variables, and successful speed-date outcomes, showing that participants who rate themselves highly are more likely to enjoy their date and be liked in return.

This confidence variate is identified by its strongest relationships, indicated as self-perceived attractiveness (attr) and social engagement (fun). The Wilks' Lambda test confirmed that this variate was significant ($p < 0.001$), and the second and third variate, while also significant by Wilks' test, explain significantly less of the outcome variate and do not improve this model (CCA Fig 1).

The Confidence variates' most influential variables were attractiveness (.86), fun (.88), sincerity (.65). (CCA Fig 2) The Outcome variates' most influential variable was the "like" answer, showing beautifully that the confidence variate explains nearly 99% of whether or not a participant likes their date, as well as the decision to match and the probability that they were liked back having strong positive relationship with the Confidence variate. The results of the

canonical correlation analysis (CCA Fig 3) show an undeniable relationship between a participant's self-esteem when approaching their speed date, and the success of the speed-dating experience.

Factor Analysis (FA)

The factor analysis determined that the cumulative variance of the factors explains 45.1% of the variation in the data, which means that 45.1% of the shared variance in the data can be explained by these four factors. Looking at the factors, we can see that they each cover a specific aspect of speed dating. An appropriate name for Factor 1 could be "Participant's Self-Perception" because all the variables relate to how the survey participant sees themselves. Factor 2 could possibly be named "Participant's Ideal Mate" because the variables relate to what the participant looks for in a partner. The third factor could be titled "Date Perception of Participant" because the variables all correspond to how the participant viewed their speed dating match. The last factor could be named "Participant Perception of Date" because the variables correspond to how the participant viewed the date on the night of the speed date. Figure FA-1 shows these factors in more detail.

The results display an interesting trend. Before the actual date, both the participant and the date find attractiveness to be the most important aspect of speed dating. However, after meeting the date, attractiveness is no longer the most important aspect of speed dating. Instead, more internal qualities such as how fun the person is and what interests the pair share. These results back up the PCA as well, with the fun rating being more important than attractiveness in determining the outcome of a speed date.

These results show that despite what speed dating participants believe before the date, attractiveness is not as important to the actual speed date event as they believe. Instead, personality aspects such as fun or shared interests are more important during the date. This shows that just being yourself and expressing your interests is important in speed dating. If you and your date share interests or your date finds you funny, then you are more likely to have a successful speed date than simply using your looks.

Correspondence Analysis (CA)

In the current study, CA revealed important information with respect to the determinants of match success in speed dating events. The analysis showed that attributes such as attractiveness, perceived personality traits, and preferred age are closely related to positive match outcomes; thus, these factors are of importance at the initial stages of romantic attraction. The Chi-square test further supported these observations by indicating that the discussed relationships are statistically significant, thus reinforcing our belief that physical appearance and age commonly play a major role in the formation of contacts during speed dating. However, it also demonstrated that some of the demographic elements, such as race and occupation, had a weaker influence on the likelihood of a match, which underlines the complexity of human attraction.

These findings are important because they challenge common views about the predictors of romantic success. While physical characteristics do matter, personality and preference predispositions can also be considered essential in determining the outcome of a speed date. This paper accentuates the complexity in human attraction and points out that perhaps these factors may interact even more in the future research with regard to long-term relationships. Moreover, the use of CA in this context opens up new avenues for analyzing and understanding the dynamics involved in dates, providing a very promising framework for future research into this area.

Lasso Regression

The final model we end with after running Lasso Regression, Forward Regression and linear regression from the output we can immediately tell that there is a strong significance and influence from the variables `exphappy`, `gender` and `date_3`. `Exphappy` is a self grade with a rating of how happy you think you will be with the people you meet at a speed dating event, `gender` was also significant showing a value of 1.12. Interestingly `date_3` is defined as have you been on a date with any of your matches which also had a strong positive coefficient of 0.77 in relation to our y variable `satis_2`. `satis_2` is a numerical value that is defined as overall how satisfied were you with the people you met, 1 being not satisfied and 10 being extremely satisfied. The variables that we ended with in the final model was variables that were also apart of all the other regression models in this question of study of lasso regression, forwards regression and linear regression.

In Lasso Figure 6 this is a visualization of the plot for best fit from the lambda coefficients of the variables. This allows us to visualize around how many variables should be outputted from lasso regression to keep a strong and informative model with variables should being between 108 and 101.

Our study shows that the final model for the research question exploring satisfaction showed some interesting outputs. We can see what has the strongest positive and negatives effects on satisfaction. Some of the interesting findings that we focused on was `exphappy`, `gender`, number of people met, similar goals and more. What we found was like the other models and research questions as well. Them being users were more satisfied if they had confidence going into the speed dating event, if the dater had already met another dater before the event, if the dater went into the event happy, we also found that users found more satisfaction when having common goals and they liked concerts. These findings were not what was originally expected to have negative effects on user satisfaction especially with the idea that if you like to go clubbing you may have less satisfaction after the speed dating event.

Ridge Regression

This analysis found internal traits, such as self- perceived fun and self-perceived intelligence, to have the largest and most positive impact on how participants rated their own attractiveness. Participants who viewed themselves as fun and intelligent were more likely to view themselves as attractive, suggesting internal perceptions play a significant role in participants' self-perceived attractiveness. Secondly, this analysis found that participants who believed that their partner placed a high value on attractiveness in return perceived themselves to be less attractive. This finding emphasizes psychological societal pressures that potentially cause self-doubt in individuals. The analysis also revealed an unexpected positive relationship between age and self-perceived attractiveness. Older participants tended to perceive themselves as more attractive suggesting increased life experience and confidence has a positive impact on how participants view themselves- contrasting with conventional ideas of attractiveness and youth.

The results of the ridge regression analysis were visualized with a coefficient plot. Positive coefficients can be seen in blue while negative coefficients can be seen in red with their respective impacts on the dependent variable, self-perceived attractiveness. As depicted in the visualization, self-perceived fun and self-perceived intelligence had the largest and most positive impact on self-perceived attractiveness while the belief in the importance of attractiveness to the partner had the largest and most negative impact on self-perceived attractiveness- further supporting the findings in this model (*see figure 4*).

The ridge regression models' findings align with other statistical techniques used in this study. Personality traits, such as fun and intelligence, seem to similarly outperform physical

appearance variables when influencing romantic connections and self-perceptions. In the context of speed dating, this suggests that internal qualities tend to be more important than external attributes. This model also reinforces CCA findings in the importance of self-perceptions. Participants with a higher level of self-confidence- reflected in self ratings of fun and intelligence- view themselves more positively and in turn positively impact their dating outcomes. Conversely, the negative influence of societal expectations found in this model may reflect a broader theme across our analyses. While not specifically found in other analyses, the pressure to conform to physical attractiveness negatively impacts the participants' confidence levels, further implicating valuable internal validations.

Limitations

There are some limitations that may have impacted this study and the results. The first issue we observed was there was a large number of missing values in the dataset. This dataset was missing so many values that 85 variables had to be removed before analysis could begin. Along with the missing variables, 75% of our data points had to be removed as well due to missing values. However, we still had a large number of variables and data points to conduct our research, so we do not believe that removing these data points had a severe negative effect on our results.

Another limitation of our data is the age of our data. This data was collected between 2002-2004 at Columbia University. Even the newest data in this dataset is 20 years old by now, so the data may be unreliable. There have been multiple breakthroughs in technology relating to dating, such as dating apps and social media, which might affect the success of speed dating in 2024. Understanding the effects of social media, dating apps, and pandemic isolation on speed dating are important aspects that are not reflected in this data.

Future Work

The best course of action to take for continuing to study the speed dating experience would be to re-run the study, but in the current day. This would allow us to compare the aspects of speed dating between the original study and now. We would also be able to add new questions that are important to the modern day that were not factors in 2004. We can add questions about social media usage, dating app usage, post-COVID social habits, and other relevant questions. Adding these questions would help us understand speed dating in a more modern aspect and would be useful for organizing new speed dating events.

f. Conclusion

Main conclusions in this study provide fascinating insights into the dynamics of speed dating by incorporating techniques in principle component analysis, factor analysis, canonical correlation analysis, lasso regression, and ridge regression. Findings suggest the importance of self-perception, personality traits, and demographic factors in a social context where rapid judgements and first impressions hold a significant influence. Specifically, positive self-perceptions of fun and intelligence reflect the significance of higher self-confidence as it relates to how participants engage with their partners and ultimately find their match. Internal qualities and compatibilities, found through the implemented analyses, emphasized and challenged previous ideals of outward physical attractions. Furthermore, demographics, such as age, play a positive and significant role in romantic connections and internal perceptions.

Additionally, there remains several practical implications that can be gathered for both speed dating organizers and future participants. Future speed dating event organizers should encourage environments that facilitate the expression of authentic personality traits during

events. They should focus on activities centered around various personality traits such as discussions that promote intelligence, fun, and interest compatibility. Confidence building workshops may help participants find their future partners and feel more assured that a match can be made. Organizers should also emphasize the importance of authenticity in their participants. Self-reflection exercises before the event will facilitate self-awareness in participants and help to improve their self-image.

Overall, these findings help speed dating participants and organizers alike navigate the complexities in an evolving world of dating. Even through a digital age evolution, these findings offer foundational insights that transcend generations and matchmaking intricacies.

g. References

- Asendorpf, J. B., Penke, L., & Back, M. D. (2011). From dating to mating and relating: Predictors of initial and long-term outcomes of speed-dating in a community sample. *European Journal of Personality*, 25(1), 16–30. <https://doi.org/10.1002/per.768>
- FINKEL, E. J., EASTWICK, P. W., & MATTHEWS, J. (2007). Speed-dating as an invaluable tool for studying Romantic attraction: A methodological primer. *Personal Relationships*, 14(1), 149–166. <https://doi.org/10.1111/j.1475-6811.2006.00146.x>
- Luo, S., & Zhang, G. (2009). What leads to romantic attraction: Similarity, reciprocity, security, or beauty? evidence from a speed-dating study. *Journal of Personality*, 77(4), 933–964. <https://doi.org/10.1111/j.1467-6494.2009.00570.x>
- Pierson, E., & Suci, A. (2015). Predicting Speed-Date Outcomes with Conversational and Network Features. Retrieved August 16, 2016,.

Figure 1
Graph: Scree Plot of Principal Components

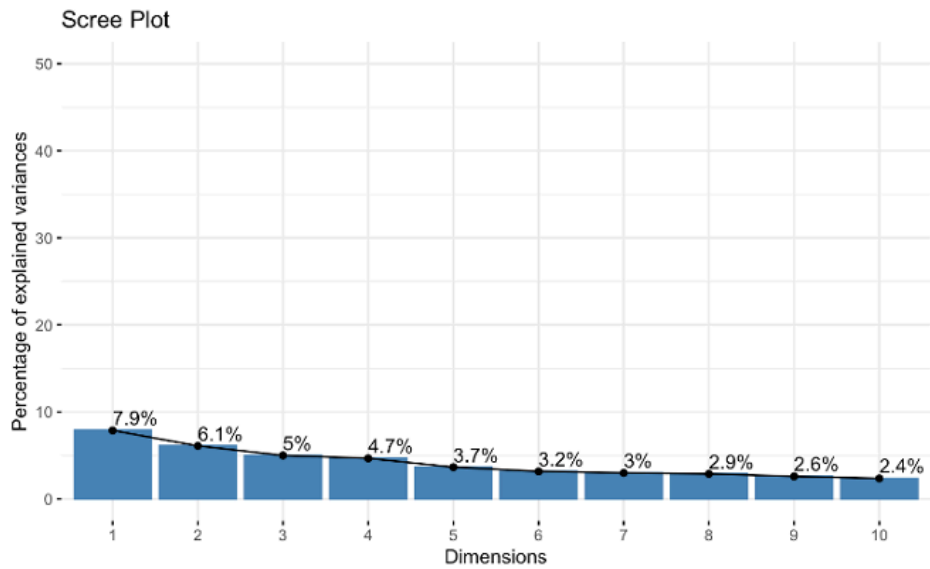


Figure 2
Graph: Canonical Loadings of Personality Traits and Dating Outcomes

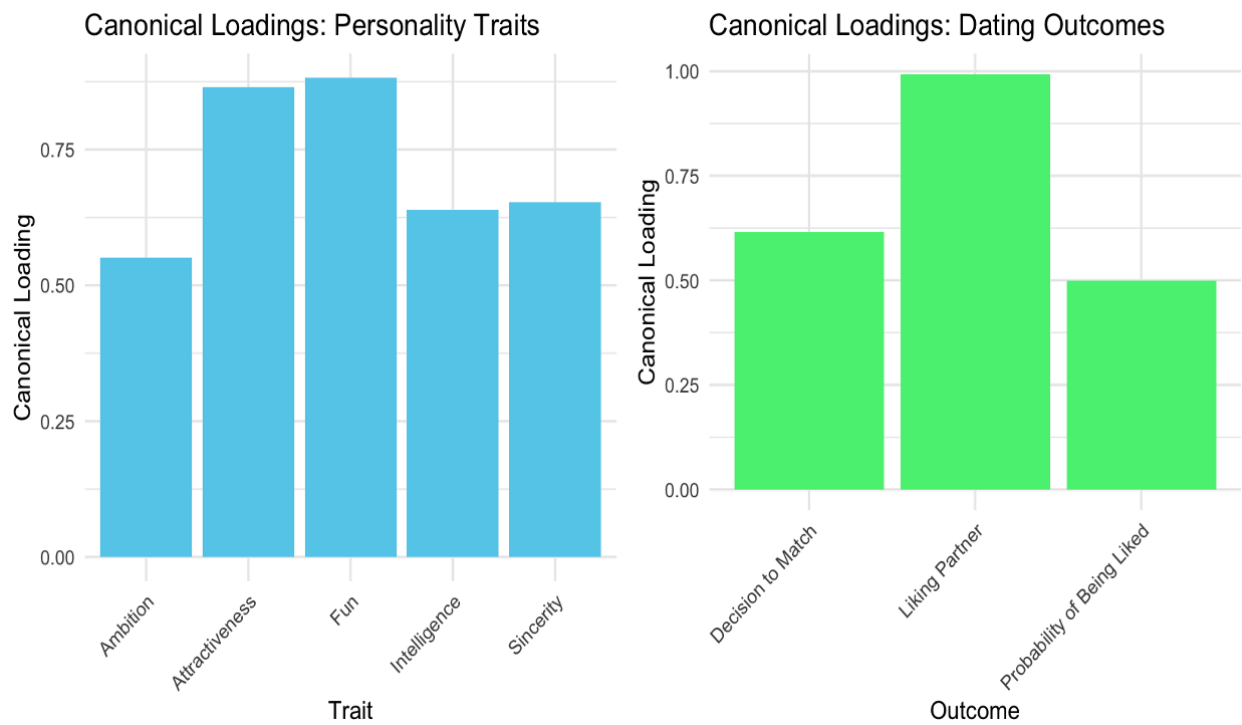


Figure 3
Graph: Confidence Levels vs Average Dating Outcomes

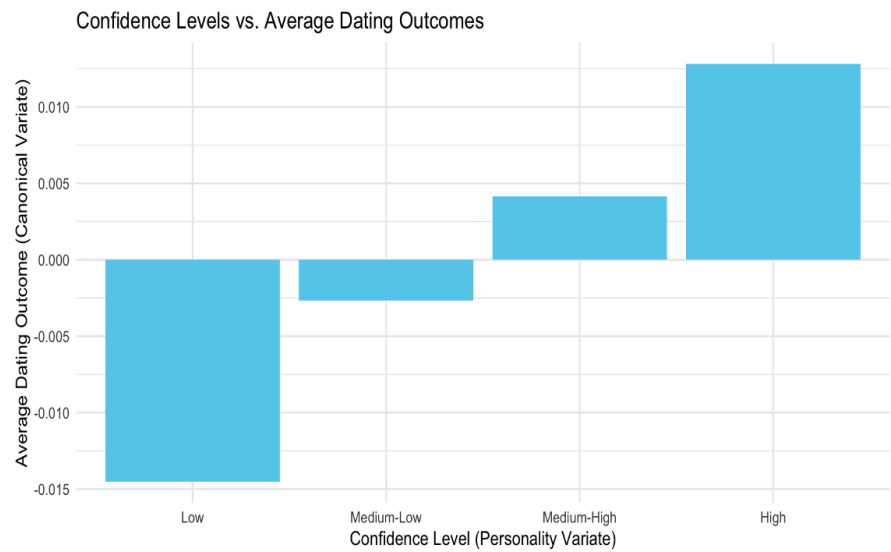


Figure 4
Graph: Factors Influencing Self-Perception of Attractiveness

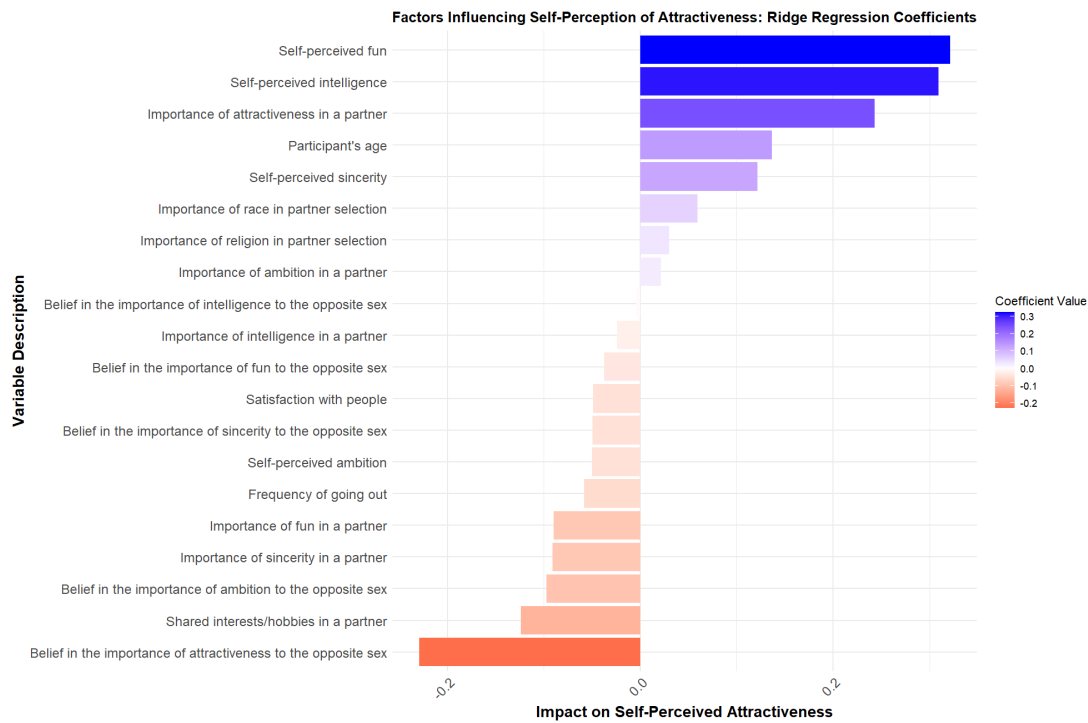


Figure 5
Graph:Factor Analysis Heat Graph with Loading Values

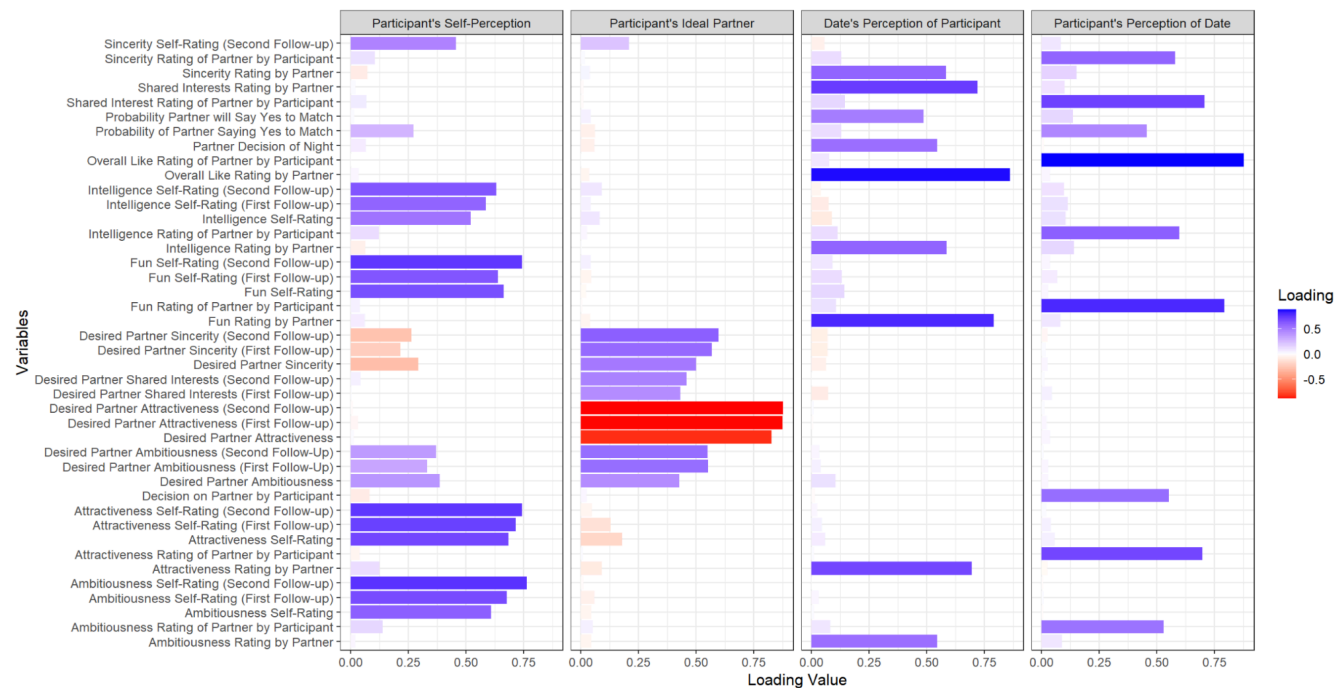


Figure 6
Graph: Lasso Regression Cross-Validation Plot

